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Azmi Haider

Research Profile

PhD candidate in Generative AI and Computer Vision at the University of Haifa.
My research focuses on generative modeling, probabilistic inference, and representation learning across computer vision, 3D reconstruction, and scientific applications.

Publications

A. Haider and D. Rosenbaum. **Direct Flow Neural Processes: Efficient Sampling via Flow Step Amortization.**
ICML 2026, Workshop on Structured Probabilistic Inference and Generative Modeling.

A. Haider and D. Rosenbaum. **Predicting 3D Structure by Latent Posterior Sampling.**
ICLR 2025, Workshop on Frontiers in Probabilistic Inference: Sampling Meets Learning.

A. Haider and D. Rosenbaum. **Tokenized Neural Fields: Structured Representations of Continuous Signals.**
NeurIPS 2025, Workshop on Structured Probabilistic Inference and Generative Modeling.

A. Haider and H. Hel-Or. **What Can We Learn from Depth Camera Sensor Noise?**
Sensors, 22(14):5448, 2022.

N. Privman-Horesh, A. Haider, and H. Hel-Or. **Forgery Detection in 3D-Sensor Images.**
CVPR 2018, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2018, pp. 1561-1569

D. Aibinder, A. Haider, and D. Rosenbaum. **Empirical Bayes Flow Matching for Continuous Cryo-EM Heterogeneity.**
Under submission (2026).

Professional Experience

2022 – 2023 | **Teaching Assistant** – University of Haifa, Israel
Led tutorials and office hours for Image Processing and Advanced Computer Vision courses, mentoring students in coursework and project development.

2021 – 2022 | **Data Science Intern (Face Recognition)** – Intel Corporation, Haifa, Israel
Developed image preprocessing, detection, and synthetic data generation pipelines for deep-learning-based face recognition systems.

2015 – 2021 | **Computer Vision Internships (3D Cameras & LiDAR)** – Intel Corporation, Haifa, Israel
Designed evaluation methodologies for 3D camera and LiDAR quality across 2D/3D sensing pipelines used in deployed computer vision systems.

Education

2022 – Present	**Ph.D. in Computer Science** – University of Haifa, Israel Thesis: *Probabilistic Inference and Generative Modeling with Tokenized Representations*. Advisor: Dr. Dan Rosenbaum. Expected graduation: August 2026.
2017 – 2020	**M.Sc. in Computer Science** – University of Haifa, Israel Thesis: *Forgery Detection in Depth Images*. Advisor: Prof. Hagit Hel-Or.
2012 – 2017	**B.Sc. in Electrical Engineering** – Technion – Israel Institute of Technology Focus: computer and software engineering.

Technical Skills

Programming: Python, PyTorch, OpenCV.

Generative Modeling: Diffusion models, flow matching, transformers, normalizing flows, variational autoencoders, autoregressive models.

Computer Vision: Implicit neural representations, 3D reconstruction, NeRF, Gaussian Splatting, probabilistic inference, representation learning.

Honors and Awards

2022 – 2025	**Excellence Scholarship** – University of Haifa
2022 – 2025	**PhD Research Fellowship** – Data Science Research Center, University of Haifa
2020	**M.Sc. graduation with honors** – University of Haifa
2011 – 2016	**Full Scholarship (Outstanding Students Program)** – Technion

Languages

Arabic — Native/Fluent
Hebrew — Fluent
English — Fluent
Spanish — Beginner